

IonQ

NYSE : IONQ · Young-Company DCF \$13.22B mkt cap · 373M sh · \$2.39B cash, zero debt · \$470M RPO (+554% YoY) at Q1 2026; SkyWater acq. (~\$1.8B cash+stock; FTC 2nd Request — unclosed)

\$35.43

THE CENTRAL QUESTION

Can IonQ capture meaningful share of a \$50B global quantum TAM by 2036 with its trapped-ion stack — or is a \$13B market cap pricing certainty Quantinuum, Google, and neutral-atom don't support?

IonQ trades at ~\$13B against a \$260-270M FY26 guide — P/S ~50x after July's sector de-rating halved the quantum complex (no adverse IonQ filing). \$2.39B cash + \$470M RPO + the \$1.8B SkyWater chip-fab deal (FTC review pending) is the optionality. Trapped-ion gives ~99.9%+ 2Q fidelity; AQ#64 hit three months early. But the competitive set is brutal: Quantinuum IPO'd at >\$14B with better logical-qubit performance, Google's Willow demonstrated verifiable advantage on chemistry, and neutral-atom scaled past 1,200 qubits. Honest math: cash-feasible raise schedules force heavy dilution before commercial inflection. Five scenarios, weighted; show your work.

PROBABILITY-WEIGHTED EXPECTED VALUE

\$9.07

expected fair value today

-74.4% vs spot \$35.43

FORWARD COMPOUNDED VALUE

+5y	+10y	+15y	+20y
\$12.99	\$18.63	\$26.74	\$38.41
0.37x spot	0.53x spot	0.75x spot	1.08x spot

WEIGHTING RATIONALE

Ultra Bear 10% Trapped-ion scaling wall + Quantinuum dominance + commercial quantum advantage delayed past 2030; equity wipes via dilution.

Bear 30% Quantinuum's IPO + better logical qubits push IonQ to #2 within trapped-ion; smaller TAM than consensus.

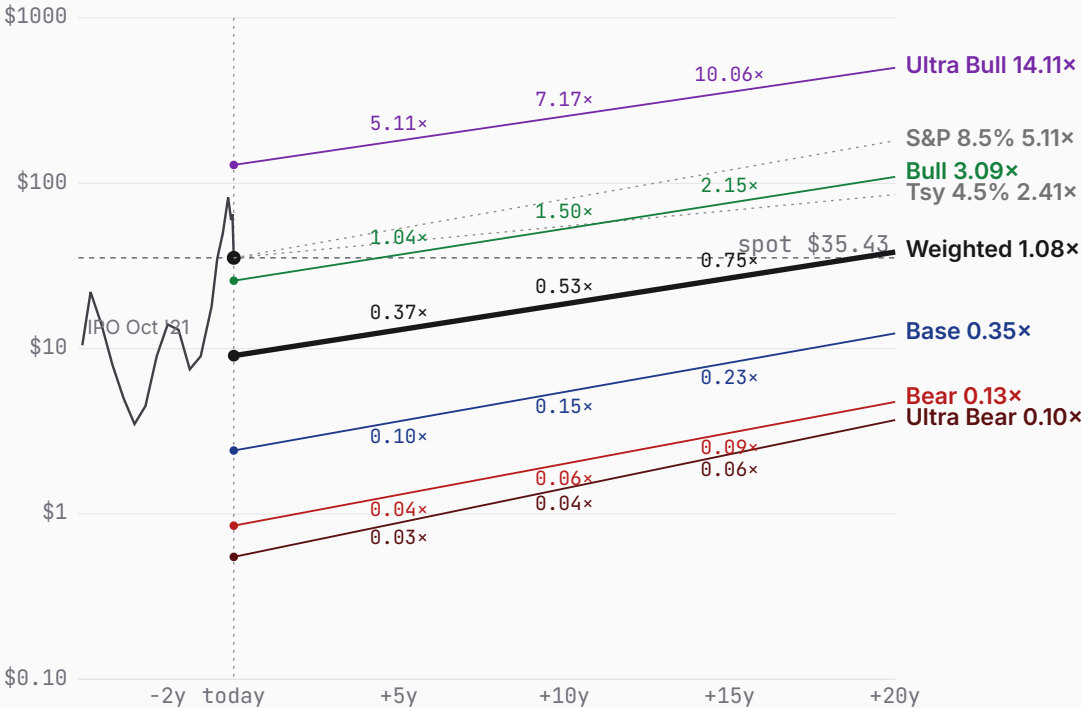
Base 42% Sequential execution on roadmap; co-leader duopoly with Quantinuum at ~8% TAM share and 17% mature margin.

Bull 15% Vertical SkyWater integration pays off; IonQ wins trapped-ion premium and platform race.

Ultra Bull 3% Tail of tails: Nvidia-of-quantum + TAM upside + competitor stumbles + multiple expansion.

Bull (15%) + Ultra Bull (3%) together contribute \$7.74 — 85% of the \$9.07 expected value despite 18% combined probability. Today's spot (\$35.43) sits 291% above the weighted expected; forward-weighted value crosses spot between +5y and +10y.

PRICE + PROJECTIONS · HISTORICAL THROUGH PROBABILITY-WEIGHTED FORWARD



ULTRA BEAR	\$0.55	BEAR	\$0.85	BASE	\$2.42	BULL	\$25.76	ULTRA BULL	\$129.21
	-98.4% vs spot		-97.6% vs spot		-93.2% vs spot		-27.3% vs spot		+264.7% vs spot
TAM 0.7% P(fail) 60% S2C 0.6 Prob 10%		TAM 2.8% P(fail) 18% S2C 1.0 Prob 30%		TAM 7.6% P(fail) 12% S2C 1.3 Prob 42%		TAM 16.0% P(fail) 8% S2C 1.5 Prob 15%		TAM 32.0% P(fail) 4% S2C 1.7 Prob 3%	
Trapped-ion sidelined; equity wiped.		Quantinuum wins; IonQ #2 niche.		Co-leader w/ Quantinuum; ~17% mgn.		Trapped-ion wins; full-stack mgn.		Nvidia-of-quantum; premium re-rate.	



IonQ scenario narratives

PROBABILITY WEIGHTS

Ultra Bear 10% Bear 30% Base 42% Bull 15% Ultra Bull 3% · Weighted expected \$9.07 (-74.4% vs spot)

ULTRA BEAR	\$0.55	BEAR	\$0.85	BASE	\$2.42	BULL	\$25.76	ULTRA BULL	\$129.21
	-98.4% vs spot		-97.6% vs spot		-93.2% vs spot		-27.3% vs spot		+264.7% vs spot

Probability 10%

Trapped-ion sidelined; equity wiped.

WHY 10%

10% reflects three compounding tail risks specific to trapped-ion: fundamental scaling ceiling around 100-500 qubits — credible physicists still debate this; Quantinuum's commercial dominance within the trapped-ion category after its >\$14B Jun 2026 IPO; the broader risk that "useful" quantum advantage stays elusive through 2030, starving the whole sector. Each individually 25-35%; conjunctive 8-12%. The FTC's pending Second Request on SkyWater adds a live blocked-deal branch to this tail.

WHAT HAPPENS

Structural failure. Quantum advantage on commercial problems takes longer than the bull-deck slides project (BCG's revised "broad advantage 2030-2040" view); the NISQ era stays a science-fair through 2030. IonQ burns \$400-600M annually to AQ#256 with no commensurate revenue ramp. Worse, trapped-ion architecture hits a fundamental scaling wall around AQ#100: ion-shuttling complexity, vacuum-chamber engineering, and laser-stability budgets all degrade nonlinearly. Quantinuum's H-series, with the same physics and Honeywell's industrial-scale manufacturing, edges ahead and consolidates the trapped-ion niche.

Meanwhile superconducting (IBM Nighthawk, Google Willow) achieves first verified commercial advantage at a price IonQ can't match per-shot. Neutral-atom scales past 2,000 atoms by FY28 and wins the cost curve. The FTC blocks the SkyWater deal, so there is no captive fab — and no fab to salvage. IonQ's \$470M RPO converts but never grows; FY30 revenue stalls below \$400M against \$500M+ cash burn. Cash exhausts mid-2029. Successive raises from \$28 down to \$2 roughly triple the share count; by FY32 IonQ is taken private at a token premium. With 60% within-scenario p_fail, distress recovery ~\$0.80. Expected per share ~\$0.55.

Probability 30%

Quantinuum wins; IonQ #2 niche.

WHY 30%

Quantinuum's completed Jun 2026 IPO — priced above range at a >\$14B valuation, \$1.68B raised — with materially better logical-qubit performance (48 logical at 99.92%) is a real competitive shift — Honeywell mfg + Cambridge Quantum SW is the most credible full-stack competitor. The broader quantum TAM consistently underperforms forecasts. IonQ executes technically but the pie is smaller and Quantinuum eats the premium slice. Within-scenario 18% p_fail reflects that even in the bear world, IonQ's cash position and DARPA contracts likely prevent bankruptcy. 30% weight reflects high-base-rate competitive entry has already happened.

WHAT HAPPENS

Execution exists but the market shape disappoints. IonQ delivers AQ#256 by 2027, but the FTC's April 2026 Second Request ends with the SkyWater acquisition blocked or abandoned — no captive fab, no 200K-qubit chip in 2028; the \$2.39B cash stays put. Commercial demand stays government- and research-heavy: DARPA QBI, DOE, NQI, a handful of pharma POCs. A Feb 2026 short report alleging most 2022-24 revenue was Pentagon-tied (contested, unconfirmed by any filing; partially offset by the MDA SHIELD and DARPA HARQ wins) is the sharpest dated evidence for that demand shape. Quantinuum's deeper logical-qubit lead wins enterprise mind-share post-IPO; IonQ becomes the credible #2 in trapped-ion, losing the big AWS/Azure/GCP enterprise-deployment race to superconducting.

Revenue ramps to ~\$1.4B FY36 — strong absolute growth but tiny vs bull-deck \$5B+. Without the fab, IonQ rents external foundry capacity and the margin ceiling drops. Operating margin matures around 8% — better than legacy services but well below cloud-software economics the current multiple implies. Cash burn averages \$300-400M through FY29; two raises (\$400M + \$1.0B) at \$30-20 prices — cash-feasible from a \$35 stock — add ~58M shares. Final ~432M. With 18% within-scenario p_fail, DCF/share ~\$0.93; expected ~\$0.85.

Probability 42%

Co-leader w/ Quantinuum; ~17% mgn.

WHY 42%

Highest single bucket because the base requires no breakthrough — sequential execution on plans in motion (Tempo shipping, SkyWater vertical integration, AQ roadmap ahead of schedule, DARPA QBI participation, RPO converting). Coexistence with Quantinuum mirrors duopoly outcomes in adjacent deep-tech markets (AMD/Intel, ASML/Nikon). 42% sits below typical 50-55% base weights because quantum has more genuine tail risk than most categories — thin middle ground between "works commercially" and "doesn't." Five- scenario structure with fatter tails forces base weight down.

WHAT HAPPENS

Modal outcome. IonQ executes the published roadmap roughly on schedule: AQ#64 done, AQ#256 by 2027, SkyWater fab chips back in 2028, the 200K-qubit-chip class reached 2028-29. Quantinuum is a credible co-leader; the trapped-ion sub-segment becomes a duopoly (~50/50 share), with IonQ winning roughly half the US government, defense, and large-enterprise quantum-cloud business via existing AWS Braket / Azure Quantum / GCP integration. Neutral-atom takes cost-sensitive optimization; superconducting takes most "deep stack" enterprise via IBM/Google brand.

By FY36 IonQ does ~\$3.8B revenue (~7.6% of \$50B TAM) at 17% mature operating margin — better than legacy hardware but below cloud-SaaS. SkyWater fab supplies internal demand plus third-party defense- electronics revenue. RPO converts steadily; multi-product customer share rises 35% → 60%+. Cash burn troughs at ~\$350M in FY28; a \$2.5B raise programme across years 1-7 at \$35-65 — starting at spot, recovering as milestones land — keeps cash≥0. SkyWater stock-portion dilution (~25M) folded in. Final shares ~451M. DCF/share ~\$2.55; with 12% within-scenario p_fail, expected ~\$2.42 — well below today's ~\$35 spot.

Probability 15%

Trapped-ion wins; full-stack mgn.

WHY 15%

Chain of conditionals: AQ#256 ~on schedule (~70%), SkyWater 200K-qubit chip delivered 2028 (~60%), IonQ — not Quantinuum — captures the trapped-ion premium position (~50%), trapped-ion beats neutral-atom and superconducting on commercially relevant workloads at scale (~50%), DARPA/DOE flows weighted to IonQ (~60%). Each individually plausible; joint ~6-8% before correlation; correlated upward to ~12-18% because conditional outcomes reinforce. 15% is a real but non-modal tail.

WHAT HAPPENS

Trapped-ion is correct architecture and SkyWater compounds. By 2028 vertical integration via SkyWater proves out: the 200K-qubit chip arrives on schedule, error correction works, and IonQ demonstrates verifiable commercial quantum advantage on a customer workload (likely chemistry or optimization for a Fortune 50). Quantinuum stays credible #2 within trapped-ion but IonQ's vertical fab + cloud-native distribution wins the platform race. Neutral-atom struggles with fault tolerance; superconducting stalls between AQ-equivalent #500 and #2000 due to error-correction overhead.

Revenue ramps to ~\$8B FY36 (~16% TAM share) with 27% mature operating margin — the SkyWater fab is now monetizing third-party customers as the dominant US trusted quantum-chip foundry. DARPA HARQ scales into a multi-billion national-security program. Federal NQI funding flows disproportionately to IonQ. Cash burn troughs FY28 then turns positive by FY30. A \$3.4B raise programme across years 1-7 at \$45-75 keeps cash≥0. Final shares ~457M. DCF/share ~\$28; with 8% within-scenario p_fail, expected ~\$26.

Probability 3%

Nvidia-of-quantum; premium re-rate.

WHY 3%

Conjunctive bull chain × TAM upside × multiple expansion × competitor collapse. Each piece individually 40-60%; joint ~3-5%. Genuine tail capturing the world where quantum becomes a category- defining tech platform and IonQ is the dominant US pure-play. The asymmetrical-moonshots tier's reason-for-being: real upside that you can't get to without it.

WHAT HAPPENS

Tail of tails. Every Bull condition hits AND the FY36 TAM is materially larger than consensus (Roots Analysis \$198B FY40 trajectory, McKinsey upper-bound \$72B by 2035 just for computing) AND IonQ achieves de facto platform monopoly in trapped-ion + becomes a meaningful chip foundry via SkyWater. Quantinuum struggles with IPO-era execution; superconducting hits fault- tolerance walls that trapped-ion's higher native fidelity sidesteps. IonQ becomes the "Nvidia of quantum" — vertically integrated, full-stack, with AWS/Azure/GCP installed customer base creating a software flywheel.

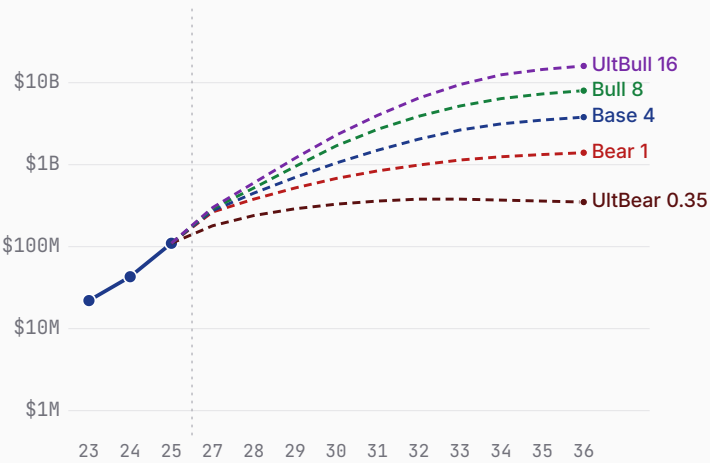
By FY36 IonQ does \$16B+ revenue at 33% mature operating margin (full-stack hardware + cloud + software-take-rate on partner apps). SkyWater fab serves the entire US national-security stack. Multiple expansion to software-hybrid premium. Modest \$3.1B raise in years 1-6 at \$55-90 maintains runway. Final shares ~441M. DCF/share ~\$134; with 4% p_fail, expected ~\$129.

IonQ business snapshot

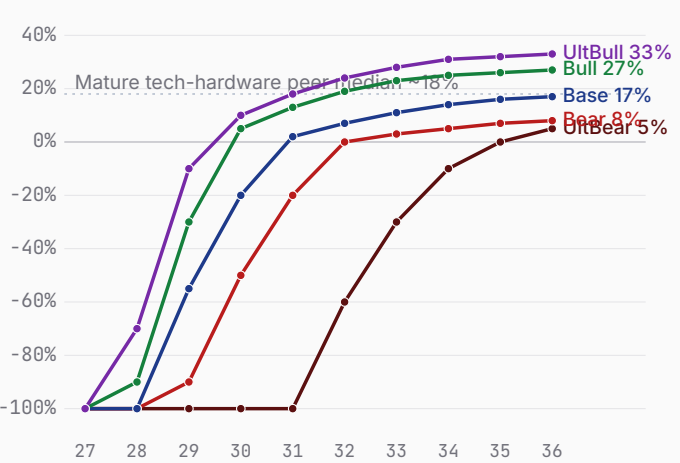
Bear \$0.85 Base \$2.42 Bull \$25.76

FY23-FY25 history + FY26 (guide) + FY27-FY36 scenario projections · fiscal years end Dec 31 · 10-K FY25, Q1 2026 10-Q, McKinsey Quantum Monitor 2026

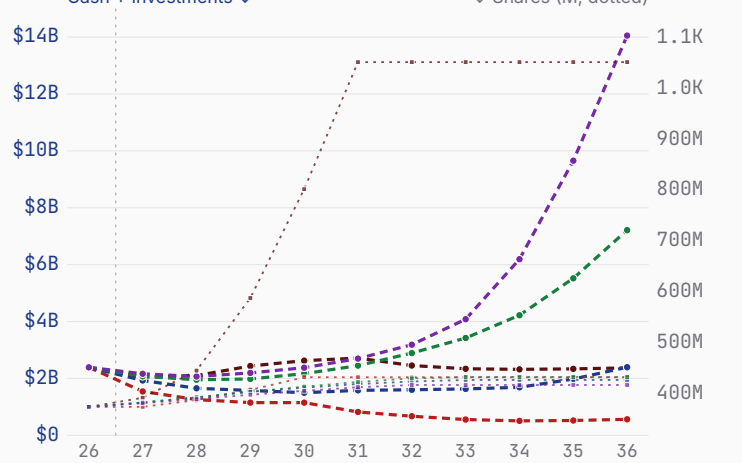
Revenue — history + scenarios to FY36 (log)



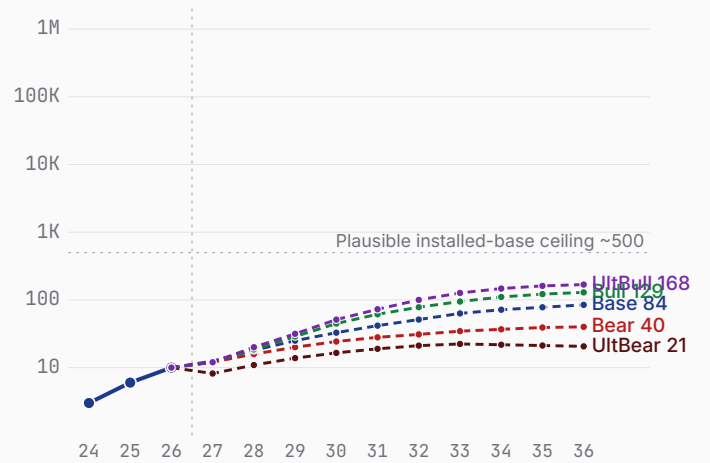
Path to profitability — op margin (FY27→FY36)



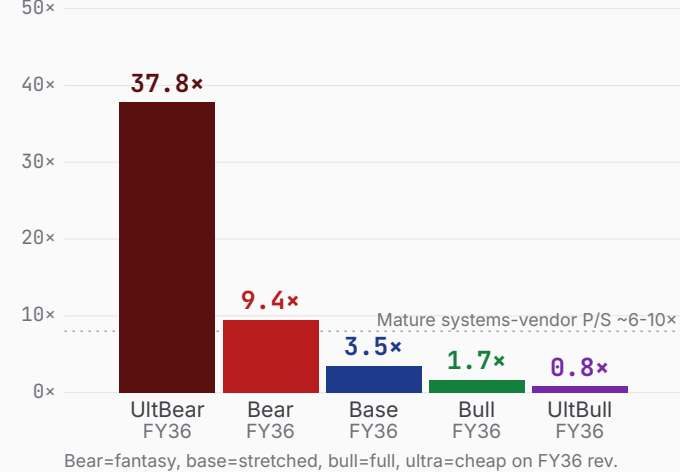
Cash + dilution (FY26 → FY36)



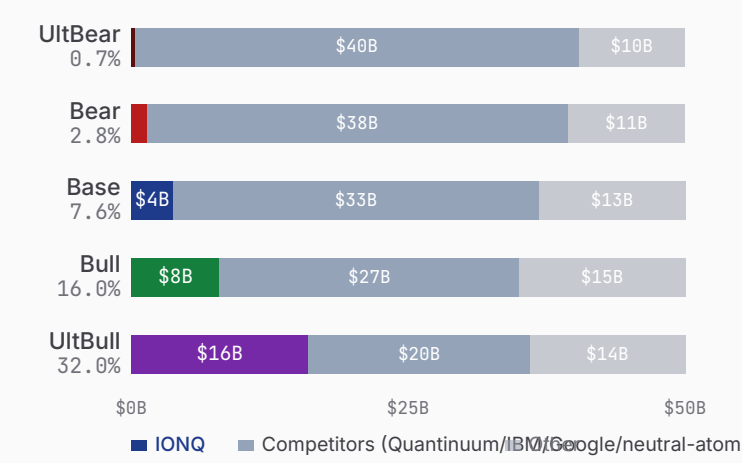
AQ#20+ class systems deployed (log)



\$13.22B mkt cap as P/S on FY36 rev



\$50B global quantum TAM (FY36) — IONQ share



Sources: IONQ Q1 2026 10-Q, FY25 10-K, IonQ Q1'26 earnings (8-K), Quantinuum S-1, McKinsey Quantum Technology Monitor 2026, BCG quantum value-creation revision, DARPA QBI Stage B selections (Nov 2025).

IonQ 7 Powers · the moat, scored

Bear \$0.85 Base \$2.42 Bull \$25.76

Arena. The race to a ~\$50B FY36 fault-tolerant quantum-computing TAM — trapped-ion (IonQ, Quantinuum) vs. superconducting (Google, IBM) vs. neutral-atom (QuEra, Pasqal, Atom Computing). **POWER ORIENTATION**
origination window: open

POWER ORIENTATION · 7 POWERS

Scale Economies	SkyWater fab vertical integration begun; not yet a cost-curve advantage.
Network Economies	No developer/data flywheel; cloud reach via AWS/Azure is rented, not owned.
Counter-Positioning	Trapped-ion stack incumbents can't pivot to without abandoning sunk superconducting fabs.
Switching Costs	\$470M RPO; algorithms compiled to IonQ's native gate set raise re-port cost.
Branding	Strong research credibility (AQ#64 early); not yet the enterprise default.
Cornered Resource · thesis leans here	Trapped-ion IP portfolio + SkyWater fab; Maryland/Duke talent lineage.
Process Power	Fidelity yield improving each generation; not yet a durable, opaque process edge.

COMPETITIVE READ

Real trapped-ion cornered-resource and counter-positioning leads, but network and scale power are unoriginated and Quantinuum's logical-qubit edge is the falsifier to watch; the origination window is open, but not indefinitely.

THE RACE · NAMED RIVALS

Quantinuum	~\$3B+ raised incl. \$1.68B IPO
~20% terminal share · Helios: 48 logical qubits @ 99.92%; >\$14B Jun '26 IPO (QNT); Honeywell-backed.	
Google Quantum AI	Alphabet balance sheet
~16% terminal share · Willow: verifiable quantum advantage demonstrated.	
IBM Quantum	IBM balance sheet
~14% terminal share · Nighthawk roadmap to 1,080 qubits; deep enterprise channel.	
Neutral-atom (QuEra/Pasqal/Atom)	\$300M+ aggregate
~10% terminal share · 1,200+ physical qubits in 18 months; different error profile.	

LEAD / LAG · FALSIFIERS

Algorithmic qubits (AQ)	64 vs 56	LEADING
Logical qubits @ >=99.9%	36 vs 48	LAGGING
Cash runway (yrs at burn)	4 vs 5	LAGGING

IonQ show your work

Bear \$0.85 Base \$2.42 Bull \$25.76 · Weighted \$9.07 (-74.4%)

ULTRA BEAR

\$0.55

BEAR

\$0.85

BASE

\$2.42

BULL

\$25.76

ULTRA BULL

\$129.21

ASSUMPTIONS

TAM Year-10 (\$B)	50
Mature share (%)	0.7
Mature op margin (%)	5
Sales-to-capital	0.6
Terminal WACC (%)	10.0
Terminal growth (%)	1.5
P(failure) (%)	60
Scenario probability (%)	10

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.18	-300%	-0.64	-0.56
FY28	0.24	-250%	-0.70	-0.54
FY29	0.29	-200%	-0.66	-0.45
FY30	0.33	-150%	-0.56	-0.34
FY31	0.36	-100%	-0.41	-0.22
FY32	0.38	-60%	-0.26	-0.12
FY33	0.38	-30%	-0.11	-0.05
FY34	0.37	-10%	-0.04	-0.01
FY35	0.36	+0%	+0.00	+0.00
FY36	0.35	+5%	+0.02	+0.01

Σ PV explicit FCF -2.28

+ PV terminal (g 1.5%) 0.07

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-2.21B
+ Cash & investments	\$2.39B
= Equity value	\$0.18B
Raised over period	\$3.50B
Dilution by FY36	+65%
FY36 shares (M)	1,052

DCF per share	\$0.17
× (1-P_fail) [40%]	\$0.07
+ P_fail × distress [60% × \$0.80]	\$0.48

= Expected per share \$0.55

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$0.89	\$1.43	\$2.30	\$3.70
0.03×	0.04×	0.06×	0.10×

ASSUMPTIONS

TAM Year-10 (\$B)	50
Mature share (%)	2.8
Mature op margin (%)	8
Sales-to-capital	1.0
Terminal WACC (%)	9.0
Terminal growth (%)	2.0
P(failure) (%)	18
Scenario probability (%)	30

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.27	-220%	-0.70	-0.61
FY28	0.38	-150%	-0.73	-0.57
FY29	0.52	-90%	-0.63	-0.43
FY30	0.68	-50%	-0.49	-0.30
FY31	0.84	-20%	-0.32	-0.18
FY32	0.99	+0%	-0.15	-0.07
FY33	1.14	+3%	-0.08	-0.03
FY34	1.25	+5%	-0.02	-0.01
FY35	1.33	+7%	+0.01	+0.01
FY36	1.40	+8%	+0.04	+0.01

Σ PV explicit FCF -2.20

+ PV terminal (g 2.0%) 0.21

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-1.99B
+ Cash & investments	\$2.39B
= Equity value	\$0.40B
Raised over period	\$1.40B
Dilution by FY36	+16%
FY36 shares (M)	432

DCF per share	\$0.93
× (1-P_fail) [82%]	\$0.76
+ P_fail × distress [18% × \$0.50]	\$0.09

= Expected per share \$0.85

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$1.31	\$2.01	\$3.10	\$4.76
0.04×	0.06×	0.09×	0.13×

ASSUMPTIONS

TAM Year-10 (\$B)	50
Mature share (%)	7.6
Mature op margin (%)	17
Sales-to-capital	1.3
Terminal WACC (%)	8.5
Terminal growth (%)	2.5
P(failure) (%)	12
Scenario probability (%)	42

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.28	-200%	-0.69	-0.61
FY28	0.45	-120%	-0.71	-0.56
FY29	0.70	-55%	-0.64	-0.45
FY30	1.05	-20%	-0.56	-0.35
FY31	1.50	+2%	-0.42	-0.24
FY32	2.05	+7%	-0.41	-0.21
FY33	2.65	+11%	-0.31	-0.14
FY34	3.15	+14%	-0.06	-0.03
FY35	3.50	+16%	+0.21	+0.08
FY36	3.80	+17%	+0.35	+0.13

Σ PV explicit FCF -2.38

+ PV terminal (g 2.5%) 2.13

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$-0.25B
+ Cash & investments	\$1.40B
= Equity value	\$1.15B
Raised over period	\$2.50B
Dilution by FY36	+21%
FY36 shares (M)	451

DCF per share	\$2.55
× (1-P_fail) [88%]	\$2.24
+ P_fail × distress [12% × \$1.50]	\$0.18

= Expected per share \$2.42

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$3.64	\$5.47	\$8.23	\$12.37
0.10×	0.15×	0.23×	0.35×

ASSUMPTIONS

TAM Year-10 (\$B)	50
Mature share (%)	16.0
Mature op margin (%)	27
Sales-to-capital	1.5
Terminal WACC (%)	7.5
Terminal growth (%)	3.0
P(failure) (%)	8
Scenario probability (%)	15

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.29	-180%	-0.68	-0.60
FY28	0.52	-90%	-0.77	-0.61
FY29	0.95	-30%	-0.79	-0.57
FY30	1.70	+5%	-0.75	-0.49
FY31	2.70	+13%	-0.72	-0.42
FY32	3.90	+19%	-0.56	-0.30
FY33	5.20	+23%	-0.10	-0.05
FY34	6.40	+25%	+0.40	+0.18
FY35	7.30	+26%	+1.00	+0.42
FY36	8.00	+27%	+1.46	+0.58

Σ PV explicit FCF -1.87

+ PV terminal (g 3.0%) 13.19

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$11.32B
+ Cash & investments	\$1.40B
= Equity value	\$12.72B
Raised over period	\$3.40B
Dilution by FY36	+22%
FY36 shares (M)	457

DCF per share	\$27.83
× (1-P_fail) [92%]	\$25.60
+ P_fail × distress [8% × \$2.00]	\$0.16

= Expected per share \$25.76

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$36.98	\$53.09	\$76.22	\$109.42
1.04×	1.50×	2.15×	3.09×

ASSUMPTIONS

TAM Year-10 (\$B)	50
Mature share (%)	32.0
Mature op margin (%)	33
Sales-to-capital	1.7
Terminal WACC (%)	7.0
Terminal growth (%)	3.5
P(failure) (%)	4
Scenario probability (%)	3

10-YEAR DCF · \$B

FY	Rev	Margin	FCF	PV FCF
FY27	0.30	-150%	-0.63	-0.56
FY28	0.60	-70%	-0.77	-0.63
FY29	1.20	-10%	-0.83	-0.61
FY30	2.30	+10%	-0.89	-0.60
FY31	4.00	+18%	-0.75	-0.47
FY32	6.50	+24%	-0.20	-0.12
FY33	9.50	+28%	+0.90	+0.47
FY34	12.50	+31%	+2.11	+1.04
FY35	14.50	+32%	+3.46	+1.59
FY36	16.00	+33%	+4.40	+1.89

Σ PV explicit FCF +2.01

+ PV terminal (g 3.5%) 55.89

EQUITY BUILD · \$B & PER SHARE

Operating EV	\$57.90B
+ Cash & investments	\$1.40B
= Equity value	\$59.30B
Raised over period	\$3.10B
Dilution by FY36	+18%
FY36 shares (M)	441

DCF per share	\$134.47
× (1-P_fail) [96%]	\$129.09
+ P_fail × distress [4% × \$3.00]	\$0.12

= Expected per share \$129.21

FUTURE FAIR VALUE · IF SCENARIO PLAYS OUT

+5y	+10y	+15y	+20y
\$181.22	\$254.18	\$356.49	\$500.00
5.11×	7.17×	10.06×	14.11×

lonQ People · Opportunity · Context · Deal

Bear \$0.85 Base \$2.42 Bull \$25.76

Underwrite the position the way a venture investor underwrites a round — across all four legs, public or private. **People** is the leg scored here (observable only); Opportunity, Context and Deal cross-reference the rest of the memo.

PEOPLE · WHO RUNS / OWNS / FUNDS IT

Niccolo de Masi1 yrs in seat

3.5%insider ownership

MEDIUMkey-person risk

Capital allocation (realized)
Niccolo de Masi became CEO in February 2025 (a director since 2021, from the SPAC-sponsor family that took lonQ public) and added the Chairman role in August 2025 when prior CEO Peter Chapman exited the board. Both academic co-founders had already left (Christopher Monroe 2023, Jungsang Kim 2024). Capital allocation is an aggressive, equity-funded acquisition roll-up: ~\$3.5B cash pro forma after a \$2B October 2025 raise, with mostly-stock deals (Oxford Ionics ~\$1.06B, SkyWater ~\$1.8B, plus Lightsynq/Qubitek/ID Quantique) driving heavy dilution. No dividend or buyback.

Incentive alignment
CEO pay is overwhelmingly equity (the exact Summary-Compensation-Table figure was not retrievable; an aggregator put it near \$90M). After ~64% say-on-pay support in 2025, the committee shifted PSU weighting toward revenue and cut maximum payouts. Insiders sold heavily in 2025 (~\$400M, including ~\$269M by the departing prior CEO and ~\$105M by de Masi); 2026 selling is smaller, mostly RSU tax withholding. Insider ownership is a derived ~3-4% (it was ~11.6% in 2024 before the share count nearly doubled).

- GOVERNANCE FLAGS
- single class of common stock, one vote per share — no super-voting, no controlling shareholder (largest holders Vanguard ~10%, BlackRock ~8%)
- combined Chair & CEO (de Masi consolidated both in August 2025) with no confirmed lead independent director since early 2025 — an independent-oversight gap
- classified / staggered board (three classes) — an entrenchment feature
- both academic co-founders departed (2023, 2024) and the prior CEO fully exited (2025) — wholesale leadership turnover
- recurring say-on-pay pushback (~64% support in 2025) and ~\$400M of 2025 insider selling
- aggressive stock-funded acquisition roll-up — repeated, material dilution

PEOPLE READ

The structural positive is genuine — one share, one vote, no controlling block, so shareholders are not disenfranchised. But this is a weak 3 on practice: a combined Chair/CEO with no confirmed lead independent director, a classified board, a ~64% say-on-pay vote, ~\$400M of 2025 insider selling, and a strategy concentrated in a CEO of under two years pursuing a heavily dilutive acquisition roll-up. Key-person risk is moderate — the science survived both founders’ departures, so it is the governance concentration, not individual dependence, that is the bigger flag.

THE FOUR LEGS

People

observable composite · 1–5

Opportunity

The scenario distribution · the business snapshot

Context

The 7 Powers analysis (Power Origination)

Deal

Open-market common — entry price weighed against the probability-weighted fair value (the -74.4% finding) + position sizing.

lonQ supporting analysis · disclaimers · glossary

Bear \$0.85 Base \$2.42 Bull \$25.76

PUSHBACK · WHY THE BASE CASE IS TOO HARSH

- 1

Best-capitalized pure-play.
\$2.4B cash + \$470M RPO + SkyWater vertical integration. Multi-year runway with optionality no other public quantum has.
- 2

AQ#64 hit early.
Tempo achieved AQ#64 three months ahead of schedule — first to do so. Real execution beats slideware.
- 3

Federal tailwind is real.
DARPA QBI Stage B + HARQ + DOE/NQI's \$2.7B/5yr authorization disproportionately benefits US-based trusted-supply players.
- 4

Trapped-ion fidelity moat.
Industry-leading 2Q gate fidelity (~99.9%+) is a physics advantage, not process — hard for superconducting/neutral-atom to close.
- 5

Cloud distribution unique.
Available on AWS Braket, Azure Quantum, Google Cloud. Only Quantinuum has comparable reach; IBM/Google don't partner cross-platform.

FALSIFICATION TRIGGERS

Bear validation

AQ#256 slips past 2027 · RPO stalls · SkyWater unclosed (FTC 2nd Request); no closing 8-K by Sep 30 '26 · Aug 5 federal-revenue detail confirms alleged gap · Quantinuum marquee enterprise win

Bull validation

AQ#256 hit by H1 2027 · First verified commercial quantum advantage on customer workload · Multi-product share >50% · 2nd large govt contract (>\$100M)

Reframe needed

If QuEra's Libra (256+ logical, Jun '26) or Pasqal (200+ by 2029) ships verified fault-tolerance ahead of trapped-ion — trips on shipped capacity, not announcement; revisit TAM share

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GLOSSARY · CONCEPTS REFERENCED IN THE NARRATIVE

Trapped-ion qubits — lonQ's qubit modality — ytterbium ions in electromagnetic traps, manipulated by lasers. Highest gate fidelity but slower clock than superconducting.

AQ (Algorithmic Qubits) — lonQ's benchmark for usable capacity. AQ#N = circuits requiring 2^N states. AQ#64 hit 2026; AQ#256 targeted 2027.

Quantum advantage — Quantum solving a problem faster/cheaper than classical. Google's Willow demonstrated verifiable advantage on chemistry Oct 2025; no commercial advantage verified yet.

Quantinuum — Honeywell-backed full-stack trapped-ion competitor. IPO'd Jun 2026 at \$60 above range (>\$14B; trades as QNT, below issue); better logical-qubit performance (48 logical at 99.92%).

SkyWater — US semiconductor foundry lonQ agreed to acquire (~\$1.8B; FTC 2nd Request Apr 2026, unclosed as of Jul 2026). Domestic chip fab — the base/bull 200K-qubit-chip path rests on it.

Tempo / Forte — lonQ system generations. Forte = current commercial (AQ#36). Tempo = next-gen, AQ#64 demonstrated, 256-qubit chip.